



AFLA BEGINNER GUIDE 1 OF 3

Getting started with AFLA

Illumina and DNBSEQ panels and exomes in EPI2ME Desktop: install, set up, first run

Education and research use only · not for clinical diagnosis or patient management

What is AFLA?

The EPI2ME experience, for short-read sequencing



Same app you know

EPI2ME Desktop runs Nanopore workflows. AFLA adds workflows for Illumina / DNBSEQ reads: pick files, click Run, read a report.



Germline and somatic

Inherited disease (single patients, trios) and tumours (tumour-only or tumour + normal). Panels and exomes.



From any starting point

Raw FASTQ, aligned BAM/CRAM, or just a VCF from a provider, Galaxy, DRAGEN or Nanopore.

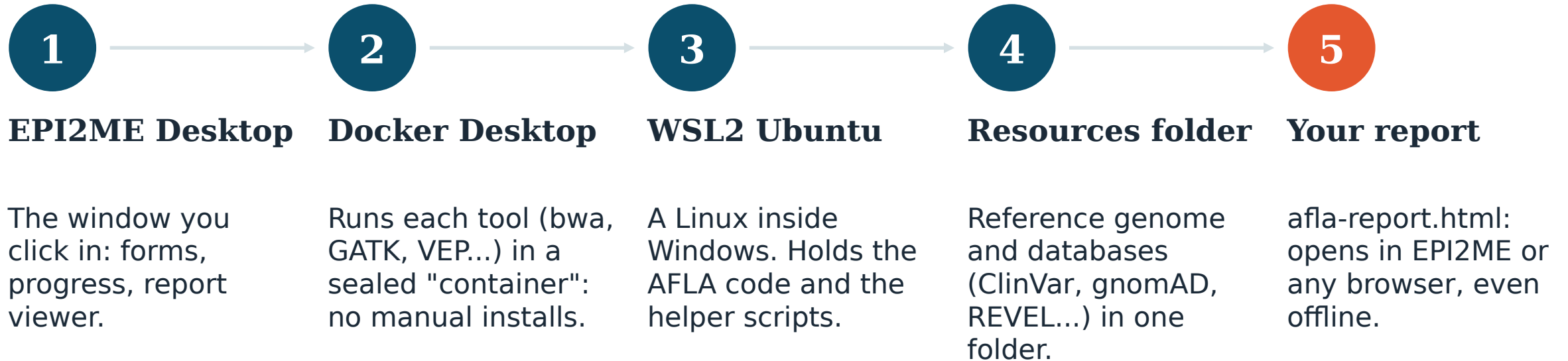


Built for teaching

Interactive report, ACMG/AMP evidence panel, exercise mode for students, printable case report.

How the pieces fit together

Four programs on one Windows laptop, and what they produce



What you need

Tested on a 16-core, 64 GB laptop; runs on smaller machines

4+

CPU cores (16 is comfortable for exomes)

12 GB

RAM minimum; 16 GB+ better. Detected automatically

60 GB

free disk for resources (25 GB with the lean option), plus your data

0

cost: all tools free; some databases free for non-commercial use

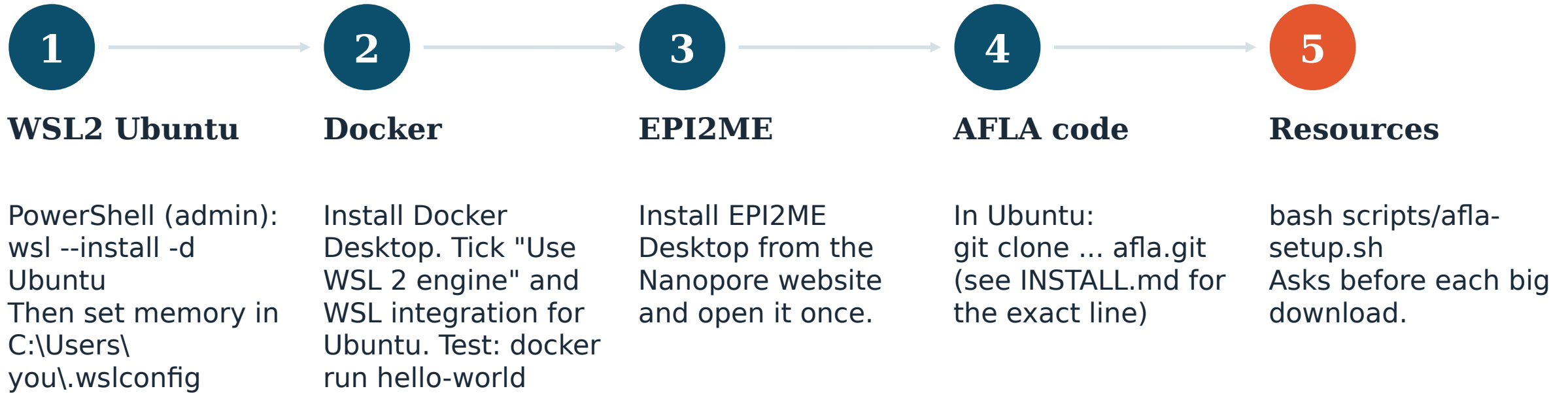
Software: Windows 10/11 (or macOS/Linux), EPI2ME Desktop, Docker Desktop, WSL2 Ubuntu.

Optional: an NVIDIA GPU (DeepVariant runs faster), Claude Code in Ubuntu as a helper.

Internet: only for installing and downloading databases; analysis runs offline.

Install in five steps

Full instructions: docs/INSTALL.md



GPU users: install only the WINDOWS NVIDIA driver, never a Linux one inside Ubuntu. Check with scripts/afla-gpu-check.sh

The resources folder

One command fills it; EPI2ME finds everything inside it

Folder	What is inside	Size
reference/	GRCh38 genome (no-alt analysis set) + bwa index	9 GB
vep/	Ensembl VEP 115 cache (genes + gnomAD v4.1 frequencies), FASTA, plugins	30 GB
clinvar/	ClinVar variants + amino-acid index for ACMG PS1/PM5	0.2 GB
revel/, alphamissense/	Missense prediction scores (non-commercial licences)	1.4 GB
constraint/, hpo/, panelapp/	Gene tolerance, phenotype ontology, expert gene panels	0.3 GB
somatic/, civic/	Mutect2 germline resource + panel of normals; CIViC cancer evidence	3.5 GB



- Re-run any time: files already there are kept.
- ClinVar refreshes when older than 60 days.
- Copy the folder to other computers on a USB disk.
- --lean skips the 30 GB VEP cache.

Put AFLA into EPI2ME

Two ways



This laptop: install script

In Ubuntu run
bash scripts/install-epi2me.sh
Restart EPI2ME: "AFLA germline" and "AFLA somatic" appear. Re-run after every update (git pull).



Any computer: import from GitHub

EPI2ME → Workflows → Import workflow → paste
github.com/fjbukhari/afla
You get "AFLA germline"; its form has an Analysis mode switch for somatic.

- Students' computers: minimum 4 cores and 12 GB RAM; CPU and memory are detected automatically.
- No Docker available? Make the VCF on usegalaxy.eu, then start AFLA from the VCF.

Where does the analysis start?

Give ONE input; AFLA skips what is already done



FASTQ folder

Raw reads (*_R1_001.fastq.gz + *_R2_001.fastq.gz).
Everything runs: QC, alignment, calling, CNV, annotation, report.



BAM / CRAM

Reads already aligned to GRCh38. Alignment is skipped; coverage, calling, CNV and report run.



VCF file

Variants from anywhere (provider, Galaxy, DRAGEN, Nanopore). Annotation, ACMG/tiers and report only.

Reference: GRCh38 only. Set "Resources folder" and the rest is found for you.

Target regions (BED): the capture kit or panel file from the vendor. Needed for CNV and TMB.

Folder names: no spaces or "+"; files are read where they are, never copied.

Which library do you have?

Library chemistry decides two switches

Library type	Examples	What to set in the form
Hybrid capture	SureSelect, Twist, IDT xGen, KAPA HyperCap	Defaults + kit BED
Tagmentation	Illumina DNA Prep (with Enrichment), Nextera Flex	Defaults + BED
Ligation	TruSeq, KAPA HyperPrep, MGI/BGI libraries	Defaults + BED
Amplicon (PCR)	AmpliSeq, TruSight/TSCA, QIAseq, CleanPlex	Tick "Amplicon panel" + give the primer BED
UMI libraries	QIAseq, xGen Prism, Twist UMI, Agilent XT HS	UMIs = inline (+ read structure) or read_name

Why amplicon mode? amplicon reads all start at the primers, so "duplicates" are real data, and primer bases must be clipped.

Why UMIs? reads from one original molecule are merged into one consensus read: errors vanish, 1-5% variants become believable.

Your first run (15 minutes)

Public practice data: the GIAB Ashkenazi trio, chromosome 20



If something goes wrong

What you see	What to do
"No paired FASTQ files found"	You chose the wrong folder: the message lists what is there. Pick the folder with the _R1/_R2 files.
"... is an index file; using ..."	Fine: you picked a .fai/.tbi; the file next to it is used.
"No bwa index found"	Use the reference from afa-setup.sh, or tick "Build bwa index" once (1 hour).
Run stops: out of memory	Close other programs; raise memory in .wslconfig; or lower "Maximum memory".
DeepVariant fails at once	GPU not visible to Docker: run afa-gpu-check.sh, or untick "Use the GPU".
Disk filling up	Delete the "work" folder of finished runs.
claude: command not found	<code>sudo ln -sf ~/.local/bin/claude /usr/local/bin/claude</code>

Words you will meet

A beginner's glossary

Term	Meaning
FASTQ	Raw reads with quality scores, straight from the sequencer
BAM / CRAM	Reads placed (aligned) on the reference genome; CRAM is the smaller format
VCF	List of variants: position, change, genotype, quality
Depth (×)	How many reads cover a base; exomes aim for $\geq 100\times$ mean, $\geq 20\times$ almost everywhere
VAF	Variant allele fraction: share of reads with the change (germline $\approx 50\%$ or 100% ; tumours lower)
gnomAD AF	How common the variant is in the population ($>1\%$ is too common for most rare diseases)
HGVS	Standard variant names: c. (cDNA) and p. (protein), e.g. c.1799T>A p.Val600Glu



Good practice from day one

- Education and research only: never report results for patient care.
- Use pseudonymous sample names (no names, no hospital numbers).
- Respect licences: REVEL, AlphaMissense, SpliceAI, COSMIC, OncoKB are non-commercial.
- Record versions: the report lists every tool and database used.
- Next: deck 2 (germline analysis) and deck 3 (somatic analysis).